

DAEN/ISEN 427 In class assignment

Due: Thu Sep 11, 2025 (in class)

Problem

In an attempt to catch the terrorists, a federal agency installs an alarm system with a surveillance camera and automatic facial recognition software to identify 10 known terrorists in a city of 1 million inhabitants (let us assume that all people present in the city are inhabitants).

The system:

- Fails to detect a terrorist 0.5% of the time (false negative rate).
- Incorrectly flags a non-terrorist 0.5% of the time (false positive rate).

A survey show that when someone triggers the alarm, people often believe with more than 99% certainty that the individual is a terrorist.

Your assignment

1. Compute the probability that a randomly selected individual is a terrorist.

Answer:

2. Calculate the probability that someone who triggered the alarm is a terrorist.

Answer:

3. If the city wanted at least a 50% chance that a flagged person is a terrorist, what maximum false positive rate would the system need?

Answer:

4. Explain why improving sensitivity (reducing false negatives) often worsens specificity (increasing false positives).

Answer:

6. Briefly discuss: Would the use of such surveillance systems likely meet legal standards? How should societies weigh this tradeoff?

For reference: Apple Face ID is designed to be inaccurate in less than 1 in 1,000,000 and Apple Touch ID to be inaccurate in less than 1 in 50,000. The theoretical probability of someone randomly guessing a four-digit passcode on the first try is 1 in 10,000. For a six-digit passcode, it is 1 in 1,000,000. In theory, this makes a six-digit passcode roughly as secure as Face ID, but there are other considerations (iPhones limit the number of failed passcode attempts, so manual brute-force attacks don't work on these devices. However, automated brute-force attacks are possible if the attacker has specialized tools).

Answer: